

1. Define $x = Q/I$, the cross section per energy loss in the close collision is given in Jackson (13.3)

$$\frac{d\sigma_{\text{close}}}{dQ} = \frac{2\pi z^2 e^4}{mv^2 Q^2} = \frac{\overbrace{2\pi z^2 e^4}^{\sigma_0}}{mv^2 I^2} \frac{I^2}{Q^2} = \frac{\sigma_0}{x^2} \quad (1)$$

This is valid for $b < d$.

For the distant collision, we have

$$\frac{d\sigma_{\text{dist}}}{dQ} = N(Q) \cdot \sigma_\gamma(Q) \quad (2)$$

where $N(Q)$ is the number of virtual quanta per unit energy interval, given by (15.58)

$$N(Q) = \frac{2}{\pi} \left(\frac{z^2 e^2}{\hbar c} \right) \left(\frac{c}{v} \right)^2 \frac{1}{Q} \ln \left(\frac{1.123 \gamma \hbar v}{Qd} \right) \quad (3)$$

With the given photoelectric cross section

$$\sigma_\gamma(Q) = \frac{8\pi^2}{137} \left(\frac{a_0}{Z} \right)^2 \left(\frac{I}{Q} \right)^3 = 8\pi^2 \alpha d^2 \frac{I^3}{Q^3} \quad (4)$$

we have for $\gamma \approx 1$,

$$\frac{d\sigma_{\text{dist}}}{dQ} = \frac{16\pi z^2 \alpha^2 c^2 d^2 I^3}{v^2 Q^4} \ln \left(\frac{1.123 \hbar v}{Qd} \right) = \sigma_0 \cdot \frac{8\alpha^2 c^2 d^2 m I}{e^4 x^4} \ln \left(\frac{1.123 \hbar v}{x I d} \right) \quad (5)$$

For a hydrogen-like atom, ionization potential is equal to the ground state energy $I = Z^2 e^4 m / 2\hbar^2$, and also $d = a_0 / Z = \hbar^2 / Z m e^2$ thus

$$\frac{d\sigma_{\text{dist}}}{dQ} = \frac{4\sigma_0}{x^4} \ln \left(\frac{1.123 \hbar v}{x I d} \right) = \frac{4\sigma_0}{x^4} \ln \left(\frac{1.123}{x} \cdot \frac{2\hbar v}{Z e^2} \right) \quad (6)$$

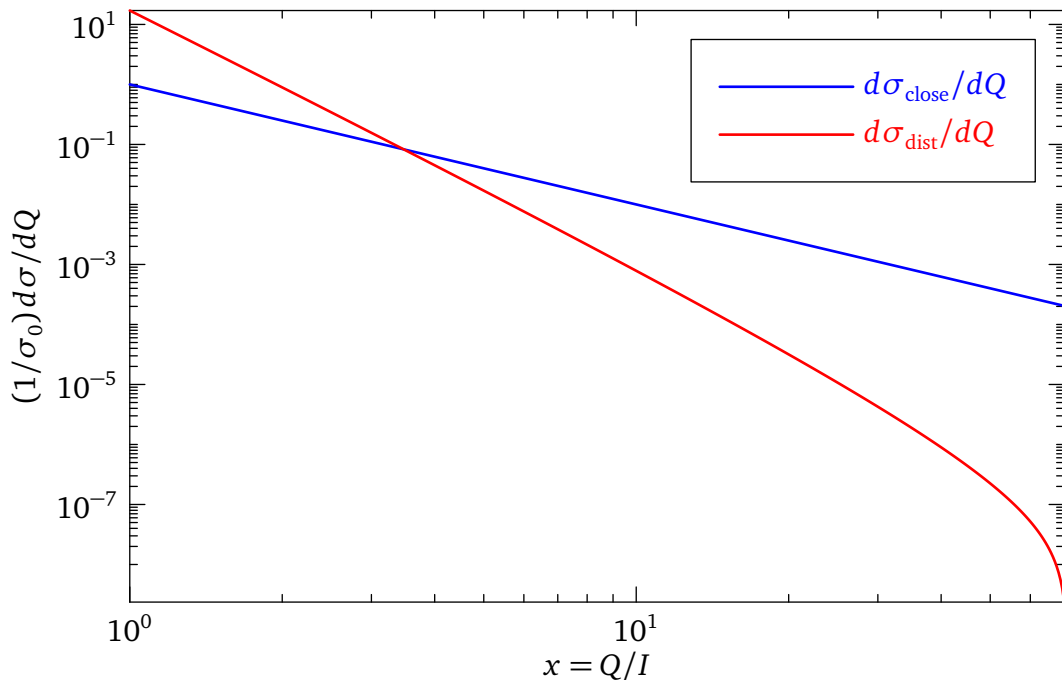
With the assumption of $I = 10^{-3} m v^2 / 2$, we see that

$$\frac{Z^2 e^4 m}{2\hbar^2} = \frac{10^{-3}}{2} m v^2 \quad \Rightarrow \quad \frac{\hbar v}{Z e^2} = \sqrt{10^3} \quad (7)$$

(6) now reads

$$\frac{d\sigma_{\text{dist}}}{dQ} \approx \frac{4\sigma_0}{x^4} \ln \left(\frac{71}{x} \right) \quad (8)$$

The two cross sections are plotted below.



2. The integrated cross section in a close collision is

$$\sigma_{\text{close}} = \int_1^{x_{\text{max}}} \frac{\sigma_0}{x^2} I dx = \sigma_0 I \left(1 - \frac{1}{x_{\text{max}}} \right) \approx \sigma_0 I \quad Q_{\text{max}} = 2mv^2 \approx 4000I \quad (9)$$

where we have evaluated the maximum energy transfer via a head-on collision of the heavy incident particle with a stationary electron.

For the distant collision, the integrated cross section is

$$\begin{aligned} \sigma_{\text{dist}} &= 4\sigma_0 I \int_1^{71} \frac{\ln(71/x)}{x^4} dx \\ &= 4\sigma_0 I \left\{ \left[-\frac{\ln(71/x)}{3x^3} \right]_1^{71} + \int_1^{71} \frac{-1}{3x^4} dx \right\} \\ &= 4\sigma_0 I \left[\frac{\ln 71}{3} - \frac{1}{3} \left(-\frac{1}{3x^3} \right) \right]_1^{71} \\ &\approx 5.2\sigma_0 I \end{aligned} \quad (10)$$

Distant collisions are ~ 5 times more frequent.

Since collision with energy transfer Q happens with rate proportional to $d\sigma/dQ$, the total energy loss is thus proportional to the integral of $Qd\sigma/dQ$ over all Q s, i.e.,

$$\begin{aligned} \Delta E_{\text{close}} &\propto \int_1^{x_{\text{max}}} \frac{xI\sigma_0}{x^2} I dx = \sigma_0 I^2 \ln x_{\text{max}} \approx 8.3\sigma_0 I^2 \\ \Delta E_{\text{dist}} &\propto 4\sigma_0 I^2 \int_1^{71} \frac{\ln(71/x)}{x^3} dx \\ &= 4\sigma_0 I^2 \left\{ \left[-\frac{\ln(71/x)}{2x^2} \right]_1^{71} + \int_1^{71} \frac{-1}{2x^3} dx \right\} \\ &= 4\sigma_0 I^2 \left[\frac{\ln 71}{2} - \frac{1}{2} \left(-\frac{1}{2x^2} \right) \right]_1^{71} \\ &\approx 7.5\sigma_0 I^2 \end{aligned} \quad (12)$$

The total energy loss is divided approximately equally between the two kinds.

From (13.14), Bethe's result is

$$\frac{dE}{dx} = 4\pi NZ \frac{z^2 e^4}{mv^2} (\ln B_q - \beta^2) \quad \text{where} \quad B_q = \frac{2\gamma^2 mv^2}{\hbar \langle \omega \rangle} \quad (13)$$

The structural similarity with Bethe's result (13.14) is evident if we take $NZ = 1$ and $I = \hbar \langle \omega \rangle$.